

A Framework for Evaluating Entity Alignment Impact on Downstream Knowledge Discovery

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Abstract. Entity alignment (EA) is a crucial process in integrating data from multiple sources, facilitating Knowledge Discovery (KD). Despite advances in EA techniques, selecting the appropriate algorithm for downstream KD tasks remains challenging due to several issues. These issues include domain entities alignment difficulties, the impact on KD tasks, and bias in data distribution. This paper presents a framework to address these challenges by providing a systematic approach to evaluate the impact of different EA algorithms based on three critical aspects: quality of alignment, information retrieved through alignment, and information imbalance or bias introduced through alignment. Our framework enables users to make informed decisions about algorithm selection, ensuring reliable, effective, and balanced KD. We demonstrate the application of the framework using a digital humanities case study, where the KD task involves enriching information about colonial collections. The choice of such a sensitive and historically imbalanced use-case allows us to highlight how the proposed framework helps identify suitable algorithms and to emphasize the importance of understanding the propagated information biases introduced through data alignment.

Keywords: Knowledge Discovery · Entity Alignment · Evaluation Framework · Digital Humanities

1 Introduction

Entity Alignment (EA) is a critical process in the domain of Linked Data, enabling the integration of distributed datasets by establishing identity links between instances that refer to the same real-world entity [22]. This process is fundamental to facilitating Knowledge Discovery across distributed datasets. *Knowledge Discovery* (KD) is often seen as an interdisciplinary process aimed at extracting useful and actionable knowledge from data repositories [5]. In the this paper, we define the KD task as finding (additional) useful and actionable information through EA across data sources. Selecting the appropriate EA algorithm for Knowledge Discovery presents significant challenges. The complexity arises from multiple factors, including the inherent difficulty of *domain entities* (entities that only show up in domain specific context [10]) alignment, the lack of well-defined KD task definition, and biases in data distribution.

EA algorithmic performance is often reported using benchmark datasets, which do not reflect the diversity of real-world scenarios [3]. The lack of ground truth in real-world cases further complicates the assessment of EA algorithm accuracy, leading to mistrust and hindering their application in fields like biomedicine [23] and historical research [8]. Additionally, the expectation of facilitating Knowledge Discovery through EA often lacks a clear definition of the desired discoveries, making it difficult to select algorithms that maximise gain in information for downstream tasks. Imbalanced node degree distribution [21] further exacerbates these challenges, as biases can lead to patterns that do not correspond to real phenomena [16]. EA algorithms that favour entities with dense neighborhoods (referred to as *head* entities) over entities with sparse neighborhoods (referred to as *tail* entities) reinforce data imbalance and introduce bias in information retrieval [10].

To address these issues, we propose a framework designed to evaluate how links introduced through different EA techniques influence downstream Knowledge Discovery tasks, specifically in terms of gain in useful information for the user. This paper presents a method that allows a user to investigate the suitability of an EA algorithm based on three features related to the task of Knowledge Discovery. In our setting, Knowledge Discovery is implemented as a set of (SPARQL) queries that matches user competency questions. EA will establish new correspondences links between entities in distributed knowledge graphs (KGs) that may lead to additional results to such queries. The suitability of an EA algorithm is determined by a) reporting the quality of the produced links; b) counting the number of additional (correct) results for specific queries, in other words the gain in information; and c) highlighting the potential information imbalance in these results introduced by these links. By quantifying these, our framework aims to guide the selection of EA algorithms that ensure balanced and accurate Knowledge Discovery, thereby enhancing the integrity and reliability of the Knowledge Discovery process. We validate our framework through a case study in digital humanities, demonstrating its potential to support critical evaluation and informed decision-making in the selection of EA techniques.

2 Background

In this section, we further define the problem based on related work and end with stating the specific contribution of this paper.

2.1 Related work

Domain Entity Alignment. Aligning *domain entities*, which only appear in a domain-specific context [10], presents significant alignment difficulties [8, 23]. The heterogeneous nature of data sources, coupled with the absence of ground truth data, creates substantial obstacles to accurate alignment. Additionally, real-world data often exhibits sparsity [21], skewness [7], ambiguity [20], and

incompleteness [18], further complicating the alignment process. Creating manually annotated ground truth data for domain entities can be counterproductive, as one of the motivations for developing automated EA systems is to address the lack of human resources for manual alignment. The manual annotation process is not only time-consuming and expensive but also prone to human error, complicating the development and validation of effective EA algorithms.

In addition, the benchmark datasets, often used to report the performance of state-of-the-art EA algorithms, typically exclude domain-specific entities [4]. This absence leads EA performance scores to inadequately reflect their effectiveness in real-world contexts. Benchmark datasets like DBpedia, Wikidata, and YAGO are constructed from structured instances from Wikipedia, where most real-world entities do not appear; even when they do, they fall under the long-tail distribution [10]. The differences in distribution between benchmark datasets and real-world datasets cause accuracy and precision scores to vary, complicating the assessment of the accuracy and quality of links produced by EA algorithms in real-world scenarios [22]. This variation can lead to potential mistrust in alignment results, hindering their application in critical fields such as biomedicine and historical research, where trust and reliability are paramount.

Lack of Task Definition. In the context of Entity Alignment, there is often an implicit expectation of facilitating Knowledge Discovery [14], yet this expectation is frequently not accompanied by a clear definition of the specific types of discoveries being sought. The absence of a well-defined information retrieval task makes it challenging to determine which EA algorithm will ultimately maximise gain in information, complicating the selection of the most appropriate algorithm for downstream Knowledge Discovery tasks. Without a formalised understanding of the desired outcomes, evaluating the effectiveness of knowledge discovery becomes infeasible. Some algorithms might excel at linking densely connected entities but perform poorly with rare or domain entities [21], leading to information bias and hindering the full potential of linked data for Knowledge Discovery. Clearly defining the goals and metrics for Knowledge Discovery by specifying the types of entities and relationships of interest and their relevant contexts is essential [2, 12]. This enables researchers to better assess the strengths and weaknesses of various EA algorithms in the context of use, leading to more informed choices that enhance the overall quality and utility of the discovered knowledge.

Bias in Data and Alignment. While the benefit of sharing knowledge is universally recognised, the potential impact of unequal knowledge distribution across datasets is less frequently critiqued. This lack of balance in information, which we refer to as bias, is generally assumed a priori when producing applications that exploit connected datasets. However, such biases can significantly impact results, leading to the emergence of patterns that do not correspond to real phenomena [6, 16].

An EA algorithm that predominantly favours aligning head entities may overlook rare or domain entities [3, 21]. This can lead to a malformed understanding

from the data, where insights and knowledge are biased towards more connected entities, masking the true diversity and complexity of the information. Selecting the right EA algorithm is therefore crucial to mitigate these biases. An algorithm that accounts for the equal distribution of information across datasets can provide a more balanced and accurate alignment, ensuring that rare entities are also appropriately linked and represented. This balance is essential for accurate Knowledge Discovery and for drawing conclusions that reflect real-world phenomena.

In certain domains, where the accuracy and completeness of data are paramount, the choice of EA algorithm can influence the outcomes of research and the validity of the insights gained. Hence, it is important to carefully evaluate and select EA algorithms that are capable of handling information biases to ensure the integrity and reliability of the Knowledge Discovery process.

2.2 Contribution of this paper

Based on these related works, we define the problem as follows. Given a set of Entity Alignment (EA) algorithms X , the challenge for the *user* is to determine which algorithm $x \in X$ is best suited for downstream Knowledge Discovery tasks. To this end, we develop a framework that consists of three criteria to guide this selection. For any given EA algorithm, the framework enables the quantification of the following criteria:

- **Entity Alignment Efficacy:** the precision, recall, and overall performance of alignment.
- **Gain in Information:** the increase in useful and actionable information resulting from the alignment.
- **Information Bias:** the extent of information imbalance introduced into the integrated dataset due to the alignment process.

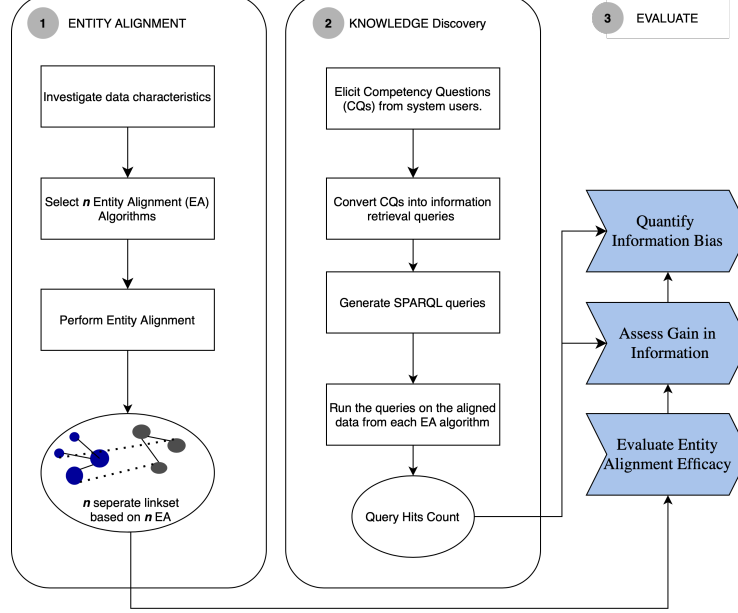
In this context, we consider two groups as potential *users* of the framework: domain researchers who might conduct linking as part of knowledge discovery for their research. The second user group consists of data managers who are responsible for managing, curating, and aligning data from one or more institutes based on domain interest and institutional strategic needs.

3 Proposed Framework

In this section, we present the proposed framework and discuss the significance of each step in quantifying Knowledge Discovery through Entity Alignment. Figure 1 outlines the complete framework.

3.1 Entity Alignment

Selecting a range of algorithms based on the characteristics of the source and target KG is crucial [12]. Such KG characteristics include skewed node degree

Fig. 1. Overview of the proposed Knowledge Discovery framework.

distribution of the entities, the presence of entities with missing attribute values, or entities with ambiguous labels. Assessing these characteristics enables users to determine which algorithms from the literature are possible candidates of EA for their specific problem.

In our framework, we recommend initially selecting multiple suitable algorithms rather than a single one. This allows users to compare the performance of different EA algorithms on the downstream KD task, ultimately facilitating the selection of the most appropriate algorithm.

Once the algorithms are selected, the user can proceed with alignment (consolidating links) based on each chosen EA algorithm. This will result in different versions of consolidated links corresponding to each algorithm. To understand the impact of these different versions of alignments on the downstream KD task, it is essential to define the task concretely. The process for constructing specific tasks for Knowledge Discovery is described in the next subsection.

3.2 Knowledge Discovery

To effectively define the Knowledge Discovery task, it is crucial to consider domain information needs [12]. One established method for capturing this is through Competency Questions (CQs) [19], which are natural language questions outline and define the scope of knowledge engineering. The advantage of using CQs in defining Knowledge Discovery tasks lies in their ability to be translated

into information retrieval queries such as SPARQL queries, which are quantifiable.

In our framework, we propose eliciting CQs directly from system users and then converting them into information retrieval queries for Knowledge Discovery tasks. The count and approximate accuracy of the hits from these queries will serve as a proxy for “gain in information” within our framework.

3.3 Evaluation

Finally, this step involves evaluating the effectiveness of Knowledge Discovery through Entity Alignment. Our framework evaluates three aspects: (1) Entity Alignment Efficacy refers to assessment of performance of the Entity Alignment algorithms, including precision, recall, and F1-score; (2) Gain in Information reports the impact of Entity Alignment on the KD process, specifically the number of new hits generated per Competency Question after alignment and the approximate accuracy of these hits; and (3) Information bias communicates whether the knowledge is skewed towards head entities or tail entities. The metrics notation and description for these assessments is given below:

Entity Alignment Efficacy We have integrated EA efficacy metrics into the framework to assess the performance of EA algorithms against ground truth data. Evaluation using standard classification-based metrics, i.e., precision, recall and F1-score, offers quantitative insights into algorithmic performance.

In cases where ground truth data is unavailable, precision (P_i) is estimated from manual evaluations on random samples from each algorithm’s alignment.

Gain in Information As previously mentioned, we count the number of new hits for each CQ after Entity Alignment as indicator of gain in information from that alignment algorithm, denoted as ΔH_{ij} . Here, i represents the algorithm and j represents the CQ. Related to this measure, we introduce AE_{ij} , representing the number of aligned entities that provide at least one answer to respective competency question (CQ_j) by algorithm i .

To estimate the accuracy of the gained information, we multiply ΔH_{ij} and AE_{ij} with the corresponding algorithmic precision (P_i) which is represented by estimated correct hits ($\widehat{\Delta H_{ij}}^{Correct}$) and estimated correct aligned entities ($\widehat{AE_j}^{Correct}$).

$$\widehat{AE_{ij}}^{Correct} = AE_{ij} \times P_i \quad (1)$$

$$\widehat{\Delta H_{ij}}^{Correct} = \Delta H_{ij} \times P_i \quad (2)$$

Information Bias To understand information bias introduced through EA in KD task, we quantify the difference between the actual new hits per CQ for aligned entities and the increase in hits per CQ if aligned entities were normally distributed in the entire dataset, denoted as the *Deviation Indicator* (DI_{ij}). To calculate this, we first introduce the expected correct number of new hits for CQ_j for algorithm i , denoted by $E(\Delta H_{ij}^{Correct})$, which is calculated by the equation below.

$$E(\Delta H_{ij}^{Correct}) = \widehat{AE}_{ij}^{Correct} \times Avg(H_j) \quad (3)$$

$Avg(H_j)$ is calculated considering if all entities (of interest) from target KG were aligned with source KG, what would be the average number of result for CQ_j . We calculate DI_{ij} as the normalized difference between the estimated correct number of hits and expected correct number of hits, using the following equation.

$$DI_{ij} = \frac{\widehat{\Delta H}_{ij}^{Correct} - E(\Delta H_{ij}^{Correct})}{E(\Delta H_{ij}^{Correct})} \quad (4)$$

The Deviation Indicator is a measure for the amount to which the retrieved (correct) counts for CQ_j deviates from the expected counts, assuming that the result for CQ in the target dataset is normally distributed. This DI_{ij} value is an indicator to what extent the retrieved results are closer to average, or shifted towards the head or tail distribution. A positive value for DI_{ij} indicates that the aligned entities contributing to the result of CQ_j is from the head distribution, while a negative value suggests the opposite.

4 Case Study

In this section, we demonstrate the application of the proposed framework through a case study focused on facilitating Knowledge Discovery in museum heritage collection data for provenance research. Provenance research [17] is a multidisciplinary field that aims to trace the ownership history and historical context of cultural objects' collection, providing insights into their historical significance. Provenance research is challenging due to incomplete historical records and the substantial volume of objects requiring thorough investigation. Here, clear understanding of decision-making processes within the system is essential to mitigate the potential biases which are often inherent in the dataset.

This study considers datasets from two Dutch museums: Wereldmuseum and Museum Bronbeek. Wereldmuseum¹ focuses on ethnographic heritage objects and their collectors, whose history goes back to reflecting colonial-era state power and expansion, while Museum Bronbeek² primarily houses collections related to military personnel and civil servants from the Royal Netherlands East Indies Army (KNIL). By aligning these datasets based on overlapping person entities

¹ formerly known as National Museum of World Culture (NMVW): <https://amsterdam.wereldmuseum.nl>

² <http://museumbronbeek.nl>

in both datasets, our KD task is to identify collectors with potential colonial backgrounds and information enrichment of the objects they collected, which is significant for domain researchers, such as provenance researchers or historians.

4.1 Data Descriptions

Wereldmuseum KG originates from heritage object collection metadata and follows the CIDOC-CRM ontology based on Linked Art recommendations. Within the context of this case study, the goal of the entity alignment is to find the same collectors who also co-exist in Bronbeek dataset. Concretely, the alignment task is to match the same individuals across KGs expressed as “actors” in various events related to objects, i.e., acquisition, transfer of custody, or production. The complete knowledge graph consists of 1,079,700,638 triples, divided into five main components: objects (976,000 entities), constituents (actors of events) (39,619 entities), historical events (70 entities), exhibitions (624 entities), and themes (438 entities). Detailed statistical distributions of these components is given in the Github repository³.

Bronbeek Museum dataset, consisting of 15,382 constituents and 94,994 related objects was extracted from the museum’s SQL database. For the purpose of this case study, it was transformed into Linked Data (LD). While we here also follow Linked Art recommendations, deviations were made to accommodate specific nuances in the dataset. The full details of the data conversion can be found in Github repository⁴.

4.2 Entity Alignment

We first analyse the dataset characteristics in terms of density, which includes the distribution of (in-)degree per (person) entity and the distribution of attribute counts per entity. In this case study for EA task, we consider the Wereldmuseum KG (39,619 person entities) as our source KG and the Bronbeek museum(15,382 person entities) entities as our target entities. Both datasets exhibit a long-tail degree distribution, meaning a small number of actors are connected to a large number of objects, while a large number of entities are connected to only one object. Another noteworthy characteristic specific to our dataset is the low number of attribute count. Only 7,500 entities from source KG have more than two attributes, while the majority of source entities possess just one attribute value, which is the entity name or label (see Github stats for support).

Given the context of the KD task and the scope of data with limited attributes along with skewed edge-degree distribution, we opted for string-based matching approaches for alignment. For this case-study, we choose simple EA

³ https://github.com/Shoilee/dh_entity_linking_v2/blob/serendipity/knowledge_discovery/stats/stats.ipynb

⁴ Github repository for all codes and documentations: https://github.com/Shoilee/dh_entity_linking_v2/tree/serendipity

techniques to maximise recall and mitigate bias towards entities with high degree distribution or high attribute count by focusing only on entity names, in other words the string literal values of their labels.

In our approach, we consider five string matching techniques for aligning person instances across datasets based on their name’s string value. 1) **Exact String Matching** is strict, focusing on precise matches, which may not always succeed due to variations in name mentions (e.g., abbreviations, full names within brackets). 2) **Initial+Surname Matching** caters to match with Wereldmuseum’s naming convention but may struggle with title identification (e.g., “Baron Haro van Hemert tot Dingshof”, where “Baron Haro” is the title) and with inconsistency in initials (e.g., “S.J. (Sjoerd) Nauta”). 3) **Surname Matching** offers flexibility but may face challenges with correct identification of complex surnames (e.g., “Baron Haro van Hemert tot Dingshof” considered as a single-part surname “Dingshof”). 4) **Fuzzy String Matching** allows for partial matches, accommodating misspellings (e.g., using the “edit distance” metric) but may lead to significant number of false positives.

Finally, 5) **DeezyMatch** [9] deep-learning based string matching technique, initially trained on place names, which we adapted for person names using the multilingual JRC-Names dataset [11]. The positive data points for training DeezyMatch were generated when the correct alternate label exhibited a match ratio above 0.6, while negative samples were created with match ratios below 0.4, aligning with the guidance in [9]. Though we utilised default hyperparameters and guidance provided by [9], by adjusting the training samples, the EA precision and recall can be tuned further.

The results of these alignment approaches were consolidated as `owl:sameAs` identity links, stored as separate linkset named graphs. We evaluate each of these strategies and these linksets in Section 4.4. Source codes are available in the GitHub repository.

4.3 Knowledge Discovery

As outlined in 3.2, we specify the Knowledge Discovery task that captures user needs. In our study, we captured user requirements for ethnographic heritage object provenance using Competency Questions (CQs) from previous research [15]. These CQs were elicited from interviews with museum professionals familiar with the domain. From the nine CQs provided, we selected five for analysis, refining their focus based on the dataset. Questions involving “which”, “what”, and rhetorical questions were transformed into counts to quantify outcomes. The resulting CQs are presented in Table 1. We then constructed SPARQL queries to answer these questions and to report gain in information. Detailed steps of this conversion is given in GitHub⁵.

⁵ https://github.com/Shoilee/dh_entity_linking_v2/blob/serendipity/README.md#knowledge-discovery

Table 1. Competency Questions and its interpretation for Knowledge Discovery Query

CQ	Query used to report result (ΔH_{ij})
CQ-1	How many actors are involved in Wereldmuseum objects’ acquisition or production events and also appear in the Bronbeek dataset?
CQ-1b	How many objects are there from the Bronbeek dataset that are collected or produced by a Wereldmuseum actor, who has a connection to at least one other object?
CQ-3	How many new pairs of actors can be formed by connecting actors through object acquisition events?
CQ-4	How many Bronbeek objects are connected to Wereldmuseum actors, where at least one (Wereldmuseum) object per actor has a known production location?
CQ-5	How many Bronbeek objects have connections with Wereldmuseum actors, where at least one (Wereldmuseum) object per actor has a connection with a historical event?
CQ-6a	How many Bronbeek objects were collected by Wereldmuseum actors and have a similar acquisition time as some of their Wereldmuseum objects?
CQ-6b	How many Bronbeek objects have unknown acquisition dates and are connected to Wereldmuseum actors, where at least one (Wereldmuseum) object per actor has a known acquisition time?
CQ-6c	How many Bronbeek objects have connections with Wereldmuseum actors, where at least one (Wereldmuseum) object per actor has a connection with a historical event and the acquisition date is similar (+/- 20 years) to other objects from the same actor?

4.4 Evaluation

Entity Alignment Efficacy In assessing the performance of the adopted EA techniques, we aim to measure precision, recall, and F1-score.

For precision, we use estimated precision based on a random sample of 50 alignment results per strategy which are manually evaluated by the authors. The unavailability of a ground truth makes it challenging to determine recall for this task. For recall, we use the performance on a different alignment task for which we do have a ground truth. This “proxy task” aligns the same source entities (persons from Wereldmuseum) to a different set of target entities (from Wikidata instead of Bronbeek), for which a ground truth consisting of 6,178 correspondences was established previously⁶. We assume that the recall results are indicative of the recall for our task.

Table 2. Recall, Precision and F1-score on Proxy Task

	instances	match	correct	Recall	Precision	F1-score
Exact Match	6178	3360	3346	0.542	0.995	0.702
Initial + Surname Match	6178	3662	3576	0.579	0.977	0.727
Surname Match	6178	8771	4991	0.808	0.569	0.668
Fuzzy String Match	6178	15028	5515	0.893	0.367	0.520
DeezyMatch	6178	3369	3359	0.544	0.997	0.704

The efficacy for selected EA strategies on the proxy task is presented in Table 2. The results are quite expected: as the string matching strategy becomes less constrained, precision decreases while recall increases. Both Fuzzy String

⁶ previously documented in the context of Wikidata:CopyClear initiative: https://www.wikidata.org/wiki/Wikidata:CopyClear/Museum_van_Wereldculturen/Canadian_creators_NMWW_not_in_MNBAQ

Match and Surname Match demonstrate good recall, with later maintaining notably lower false positives compared to Fuzzy Match. Interestingly, DeezyMatch shows performance similar to Exact Match. This could be attributed to the strict criteria used for training samples (positive sample with similarity score ≥ 0.6 and negative samples having score ≤ 0.4), favoring higher accuracy over recall.

Table 3. Performance evaluation on the Wereldmuseum-Bronbeek alignment task based on 50 different random samples (selected separately from each algorithms’ consolidated links)

	Total Matched	True Positive	Precision	(Proxy) Recall	F1-score
Exact Match	351	50	1	0.542	0.703
Initial+Surname Match	978	46	0.92	0.579	0.711
Surname Match	51376	9	0.18	0.808	0.294
Fuzzy String Match	3533880	0	0	0.893	0.000
DeezyMatch	318	49	0.98	0.544	0.700

In the manual evaluation of random correspondence samples, we assigned scores ranging from 0 (definitely false) to 4 (definitely correct). *Precision* was calculated by counting any score above 1 as true-match. The results are presented in Table 3. The precision scores for Exact Match, Initial+Surname Match and DeezyMatch are similar to the ones in the proxy task. However, the precision for Surname Match and Fuzzy Match are quite different due to the source-target match ratio. So, we can conclude that the precision score from the proxy task do not translate well to our actual case-study task.

As previously mentioned, to estimate recall in the current alignment task, we take the respective EA strategies’ recall score from the proxy task. We also use this recall to calculate F1-score for the random sample. Due to the lower precision score in random sample the F1-score suffers for Surname Match and Fuzzy String Match, indicating the remaining three EA strategies could be a better match for current use-case. As Fuzzy Match demonstrated zero precision in random samples, we exclude this algorithm from further consideration.

Gain in Information To quantify gain in information on the downstream KD task, we report on the count generated by each EA strategy per CQ as well as their estimated accuracy through the metrics presented in Section 3.3.

We utilise the questions from Table 1, and corresponding SPARQL queries to report ΔH_{ij} . We also report AE_{ij} (how many matched entities contributed to these hits). By comparing ΔH_{ij} among results yielded using different alignment strategy informing us how much information we gained.

As depicted in Table 4, the columns AE_{ij} and ΔH_{ij} demonstrate that alignment strategies based on relaxed constraints such as Surname match yield more results compared to those based on strict constraints (Exact match and Deezy-Match match). We also can see that as the complexity of the CQ increases due to the addition of more filtering criteria imposed by the query, some strict matching strategies may fail to produce any results and strategies based on relaxed

constraints prove beneficial as they still generate some results. Interestingly, despite stricter matching criteria Initial+Surname match could manage to produce result for all CQ, except one.

To gain insight of the estimated accuracy of KD task, we report on $\widehat{AE}_{ij}^{Correct}$ and $\widehat{\Delta H}_{ij}^{Correct}$ as mentioned in Section 3.3. When comparing the estimated correct hits between Surname Match and Initial+Surname Match for certain CQs (specifically CQ-5, CQ-6a, and CQ-6c), it is notable that the correct results generated by Surname Match may be fewer than those generated by Initial+Surname Match. This occurs even though Surname Match initially aligned significantly more entities than Initial+Surname Match.

Table 4. Performance Evaluation of EA for downstream task using the framework metrics from Section 3.3. The Query Path column shows the (simplified) path in the KG related to CQ. Here, $\Rightarrow$ represents the established link through alignment.

	Query Path	Entities		Hits		Accuracy	Bias
CQ	Source $\Rightarrow$ Target	AE_{ij}	$\widehat{AE}_{ij}^{Correct}$	ΔH_{ij}	$\widehat{\Delta H}_{ij}^{Correct}$	$E(\Delta H_{ij}^{Correct})$	DI_{ij}
DEEZY MATCH							
CQ-1	actor $\Rightarrow$ actor	87	85.26	87	85	N/A	N/A
CQ-1b	object $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	36	35.28	325	319	423	-0.248
CQ-3	actor $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ actor	2	1.96	2	2	N/A	N/A
CQ-4	place $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	27	26.46	240	235	318	-0.259
CQ-5	event $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	0	0	0	0	0	0.000
CQ-6a	time $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ time	0	0	0	0	0	0.000
CQ-6b	time $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	0	0	0	0	0	0.000
CQ-6c	place $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object $\rightarrow$ time	0	0	0	0	0	0.000
Avg.							-0.072
EXACT MATCH							
CQ-1	actor $\Rightarrow$ actor	117	117	117	117	N/A	N/A
CQ-1b	object $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	36	36	707	707	432	0.637
CQ-3	actor $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ actor	0	0	0	0	N/A	N/A
CQ-4	place $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	31	31	0	0	372	-1.000
CQ-5	event $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	2	2	20	20	24	-0.167
CQ-6a	time $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ time	0	0	0	0	0	0.000
CQ-6b	time $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	0	0	0	0	0	0.000
CQ-6c	place $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object $\rightarrow$ time	0	0	0	0	0	0.000
Avg.							-0.076
INITIAL+SURNAME MATCH							
CQ-1	actor $\Rightarrow$ actor	277	255	277	255	N/A	N/A
CQ-1b	object $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	116	107	2,153	1,981	1,281	0.547
CQ-3	actor $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ actor	9	8	10	0	N/A	N/A
CQ-4	place $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	83	76	1,841	1,694	916	0.848
CQ-5	event $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	11	10	312	287	121	1.364
CQ-6a	time $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ time	1	1	16	15	17	-0.111
CQ-6b	time $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	0	0	0	0	0	0.000
CQ-6c	place $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object $\rightarrow$ time	1	1	14	13	17	-0.222
Avg.							0.346
SURNAME MATCH							
CQ-1	actor $\Rightarrow$ actor	2,680	482	2,680	482	N/A	N/A
CQ-1b	object $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	1,905	343	94,298	16,974	4,115	3.125
CQ-3	actor $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ actor	392	71	975	176	N/A	N/A
CQ-4	place $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	507	91	12,776	2,300	1,095	1.100
CQ-5	event $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	19	3	912	164	41	3.000
CQ-6a	time $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ time	3	1	23	4	10	-0.574
CQ-6b	time $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object	7	1	163	29	15	0.940
CQ-6c	place $\rightarrow$ actor $\Rightarrow$ actor $\leftarrow$ object $\rightarrow$ time	3	1	21	4	10	-0.611
Avg.							0.997

Information Bias The Deviation Indicator (DI_{ij}), calculated using Equation 4, assesses whether the information obtained through alignment is normally distributed for the intended entities. For example, if a CQ asks, "How many objects are there in the Bronbeek dataset under specific conditions?", it seeks to enrich information about objects' from the target dataset by aligning actors'. DI_{ij} measures the extent of second-order information bias introduced, comparing the actual hits to the average hits ($Avg(H_j)$) expected if all target entities were aligned. $Avg(H_j)$ is calculated based on entities on the right side of the ' $\Rightarrow$ ' in the Query Path column in Table 4. For CQ-1b, CQ-4, CQ-5, and CQ-6b, $Avg(H_j)$ is the average number of objects per actor (12); for CQ-6a and CQ-6c, it is the average time information available per actor (18). For CQ-1 and CQ-3, average calculation is not applicable as they enrich actor information, hence no second-order information enrichment was achieved.

The last column in Table 4 lists the individual values of DI_{ij} for each CQ. This shows that indeed for some CQ-EA combinations, the actual hits are shifted toward *head* (positive score) while for other more towards *tail* (negative score). We also report on the average DI_{ij} for all CQs. We see that Exact Match and DeezyMatch exhibit the least amount of information bias, in fact they are leaning more towards the *tail*; note that, they also produce zero results for some CQ. We see that for both Initial+Surname Match and Surname Match, the average values for DI_{ij} are positive. This indicates that in general, these information produced through these EAs are more skewed than expected. This points at potential introduction of knowledge imbalance or bias through alignment. Note that, to generate a positive score not all the CQs generated skewed outcome (for example, CQ-6c). Surname Match exhibits more bias than the others, implying that allowing more false positives exacerbate existing data bias.

4.5 Summary

Following the framework, we quantified the quality of the produced links, their utility for the Knowledge Discovery task, and the degree of information imbalance introduced to the source dataset from the target data. In this case-study, ensuring trust and fair alignment is paramount. Therefore, for the given task, an optimal choice of alignment could be the Initial+Surname Match out of the initially selected five strategies. Despite its simplicity and potential to overlook false negatives, it manages to generate result for all CQs except one. Moreover, in sensitive cases such as provenance research—aimed at addressing past injustices—this EA with more precision may be preferable. However, this EA involves information bias for more CQ, which should be clearly communicated to the system user.

5 Discussion

Our case study demonstrates that successful alignment does not necessarily ensure an equal distribution of Knowledge Discovery opportunities across all entities. It is crucial to evaluate whether aligning different data sources genuinely

results in gain in information. As evidenced by our Exact Match and Deezy-Match methods, although they achieve alignment, they often fail to answer many questions of interest to domain users. On the other hand, emphasising recall to maximise gain in information may introduce information bias through false alignments, as we have seen in the case of Surname Match and to some extent in Initial+Surname Match. Therefore, while striving to gain more knowledge, it is critical to understand whether the Knowledge Discovery opportunities are equitably distributed across the dataset or if they are merely enriching information for entities that have been historically favoured.

The primary benefit of this work is its capacity to quantify critical measures, thereby supporting transparency to the generated outcomes within a system where Knowledge Discovery is achieved through data alignment. By quantifying three dimensions—link quality, gain in information, and information bias—the proposed framework equips users with the necessary information to select the most suitable algorithm. Furthermore, in scenarios where maximising gain in information over minimising bias is favoured, data managers can explicitly communicate the extent of information imbalance introduced through this framework; this practice is coherent to tool criticism practices [1]. This explicit communication helps mitigate undesired outcomes in potential Knowledge Discovery tasks by making information imbalances in real-world knowledge graphs more visible.

While we validate the framework through a specific case study, several limitations must be acknowledged. First of all, there is a clear need to explore more advanced EA algorithms within our framework. It would be interesting to incorporate techniques such as embedding-based EA methods to analyse the interplay on alignment accuracy and biases, particularly in complex datasets with domain entities, such as this one. Nevertheless, as long as such methods produce correspondences, our framework can be applied on any sophisticated EA techniques. For the current paper, we focus on the framework and validation through simple matching strategies to demonstrate it.

The types of bias discussed in this work primarily focus on the structural biases [3] present in datasets, such as imbalanced node degree distributions. Other forms of bias, including race, gender [6], and socioeconomic disparities [13], are not explored in detail here. Though previous works already attempted these issues, these dimensions of bias yet represent significant avenues for future research, as they play a crucial role in ensuring fairness and inclusively in Knowledge Discovery systems. Investigating these biases could broaden the understanding of how alignment algorithms impact various societal groups and improve the overall fairness of data-driven systems.

Next to that, developing strategies to create or approximate ground truth data for EA evaluations will be beneficial. Methods such as crowd-sourcing annotations or leveraging domain-specific heuristics could provide valuable benchmarks for assessing algorithm performance in the absence of existing ground truths. Lastly, but most importantly, expanding the application of our framework to other domains beyond provenance research will be essential for validating its broader utility. By testing the framework in areas such as bio-medicine, his-

torical research, and other fields that rely heavily on data quality, we can refine and adapt our approach to meet a wider range of Knowledge Discovery needs. Here, extensive user studies can give insights into the understandability of the metrics, as well as user preferences towards the weighting of the various metrics to come to a balanced selection.

6 Conclusion

In this paper, we presented a framework for evaluating Entity Alignment (EA) algorithms in the context of the Knowledge Discovery task. Our framework employs a three-tier evaluation approach that quantifies the quality of alignment, the extent of gain in information, and the degree of information imbalance introduced by the links. The ability to quantify the impact of EA on the Knowledge Discovery task provides users with the insights needed to make informed decisions. This capability is crucial for building trust in the results generated by EA systems and ensuring that Knowledge Discovery is effective and equitable. Through a case study focused on provenance research in museum heritage collections, we demonstrated the practical application of our framework.

In conclusion, our framework offers a foundation for enhancing the effectiveness of EA algorithms in multi-source data integration contexts. By continuing to refine and expand this approach, we aim to enhance the reliability, trustworthiness, and utility of Entity Alignment, ultimately contributing to the advancement of various fields that depend on integrated data.

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